The normalization condition determines the parameter λ, whereas β and hence the temperature arises from the choice of average energy, i.e. from solving the equation

$$\frac{\int H(z)e^{-\beta H(z)}\mathrm{d}\omega}{\int e^{-\beta H(z)}\mathrm{d}\omega} = \bar{\mathrm{E}}$$

Alternatively, we may take the position that the temperature is specified and the mean energy is a direct consequence of this choice.

This defines the phase space density of the canonical ensemble which we refer to as the Gibbs-Boltzmann density:

$$\rho_\beta = Z_{\mathrm{can}}^{-1} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}. \tag{6.3}$$

where the parameter $\beta = 1/(\mathrm{k_B T})$. The quantity Z_{can} is a normalizing constant so that this is a probability density. The canonical ensemble is referred to as the NVT ensemble, due to the a constancy of the number of particles, volume, and temperature.

Now let us see the consequences of assuming that the states of a given system are canonically distributed. In order for the density in (6.3) to be normalizable, it is necessary that the quantity Z_{can} defined by

$$Z_{\mathrm{can}} = \int e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\mathrm{d}\omega$$

be finite, i.e., that the integral converge. Here the form of the system and the assumptions regarding boundaries come into play. If we have a standard molecular Hamiltonian $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ with $\mathrm{N_c} = \mathrm{N_d}$ degrees of freedom, then the integral separates

$$Z_{\mathrm{can}} = \int_{\mathcal{D}} e^{-\beta H(\boldsymbol{q},\boldsymbol{p})}\mathrm{d}\omega = \int_{\mathbb{R}^{\mathrm{N_c}}} e^{-\beta \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2}\mathrm{d}\omega_p \times \int_{\mathcal{Q}} e^{-\beta U(\boldsymbol{q})}\mathrm{d}\omega_q,$$

where $\mathcal{Q}$ is the accessible position domain, $\mathrm{d}\omega_p = \mathrm{d}p_1\mathrm{d}p_2\ldots\mathrm{d}p_{\mathrm{N_c}}$ and $\mathrm{d}\omega_q = \mathrm{d}q_1\mathrm{d}q_2\ldots\mathrm{d}q_{\mathrm{N_c}}$. If U is smooth and the domain $\mathcal{Q}$ is finite, then clearly both integrals are well defined, so the normalization is sensible. When U has singularities, it is important that at these points $U \to +\infty$ instead of $-\infty$. Moreover, if no boundary conditions are employed to restrict the particles, we would need to introduce an additional restriction on the potential U in order to ensure that the configurational integral converges.

Typically one assumes that (i) U is smooth except on a low-dimensional set of singular points, (ii) U is bounded below (which is normally the case if a potential such as Lennard-Jones, steeply repulsive at short range, is incorporated between all pairs of atoms), and (iii) U grows sufficiently rapidly as $\boldsymbol{q} \to \infty$, in the case of an infinite domain.

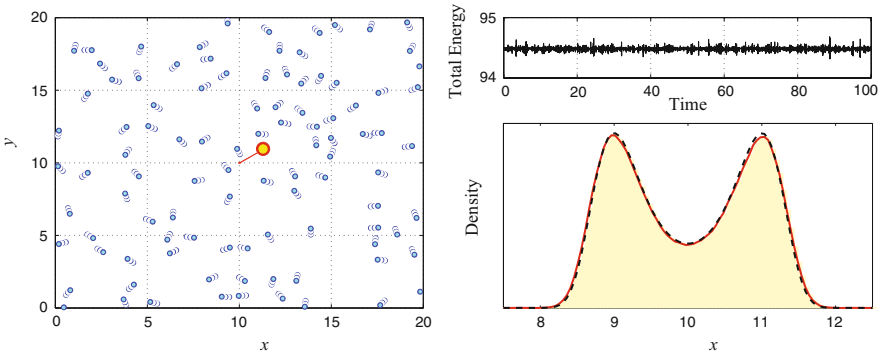


Fig. 6.2 We simulate 100 particles in the plane (*left*) using a periodic box of length $L = 20$, with the Lennard-Jones potential governing all pairwise interactions. One of these particles, marked in *red*, also feels a double-well potential in both coordinates. The total system energy (plotted in the *top right*) is conserved (with some variance due to numerical error). The distribution for the x-coordinate of the *red* particle (plotted in the *bottom right*), averaged over a single long trajectory (*solid red*) shows good agreement with the expected canonically distributed result (*dashed black*)

For example a collection of atoms on the infinite domain interacting in Lennard-Jones potentials does not satisfy the third condition, and it does not have a well defined canonical partition function. We can understand this by considering what happens over time in such a system. For simplicity, just take the case of a one degree-of-freedom system with $H = p^2/2 + q^{-12} - q^{-6}$, $q > 0$. If the temperature is low, the particle will remain for a long time in the vicinity of the well bottom. At any given time there is a positive probability of finding the particle essentially any distance from the origin, although the probability of being found very near the origin is extremely small due to the repulsive steep wall. Due to the presence of the thermal bath, eventually the particle will receive enough energy to overcome the attractive force and will be ejected and tend to infinity. The restriction to a bounded domain thus eliminates this problem which otherwise would complicate the technical treatment.

In order to use a one degree-of-freedom model on an infinite domain in the canonical setting, we would need to incorporate a restraining potential, slightly modifying for example the Lennard-Jones potential to $U(q) = q^{-12} - q^{-6} + \varepsilon q^2$ (Fig. 6.2).

6.1.4 The Canonical Density and Thermodynamic Averages

The distributional approach described above and in the previous chapter and used to define microcanonical averages on a bounded energy surface must be modified to make it applicable to the canonical ensemble. Recall that in the microcanonical setting we assumed ρ was a smooth density defined on the bounded energy surface

Σ_{E} or, more generally, a distribution. For a canonical system there is no longer a constraint of the form $H = \mathrm{E}$ and thus we must allow the phase space to include essentially all points for which the position coordinates lie in the defined configurational domain (a 3-torus, if periodic boundary conditions are used). In general the momenta are unbounded. It thus becomes necessary to restrict the set of functions on which the functional $\langle \cdot, \rho \rangle$ acts. In the case of periodic boundary conditions we will assume that, for any observable of interest,

$$|\phi(\boldsymbol{q},\boldsymbol{p})| = \mathcal{O}(\|\boldsymbol{p}\|^{\alpha}), \qquad \boldsymbol{p} \to \infty,$$

for some positive constant α.

If the position domain $\mathcal{Q}$ is unbounded (for example choosing $\mathcal{Q} = \mathbb{R}^{\mathrm{N_c}}$) then some other restriction is needed on the potential energy function U (and on the observables) in order that averages are well defined. Typical assumptions for potentials on the unbounded domain are that (i) U is bounded below, and (ii) U grows sufficiently rapidly as $\boldsymbol{q} \to \infty$. This ensures that

$$\int_{\mathcal{Q}} \exp(-\beta U(\boldsymbol{q}))\mathrm{d}\omega_q < \infty.$$

Under assumptions on the potential energy as discussed above, and with ϕ continuous and allowed to grow at most polynomially fast at infinity, we may proceed to define averages with respect to the canonical density ρ_β, namely

$$\mathrm{Av}_\beta(\phi) = Z_{\mathrm{can}}^{-1} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega. \tag{6.4}$$

6.1.5 Calculation of Temperature

Consider the one degree-of-freedom model $H(q,p) = \frac{p^2}{2m} + U(q)$ on the unbounded domain, with U bounded below and growing sufficiently rapidly as $|q| \to \infty$. The associated canonical density is

$$\rho_\beta = Z_{\mathrm{can}}^{-1} \exp\left(-\beta\left[\frac{p^2}{2m} + U(q)\right]\right),$$

where

$$Z_{\mathrm{can}}^{-1} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\left(-\beta\left[\frac{p^2}{2m} + U(q)\right]\right)\mathrm{d}p\mathrm{d}q.$$